



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, SRI CITY

MID EXAMINATION – SPRING 2026

Speech processing

CSE:UG3(PE)

Date: 24-02-2026

Duration: 90 Mins (09:30-11:00PM)

Max. Marks: 20

Instructions:

Roll No: _____

1. Answer the following. 4x5M=20M

1	Given a discrete time signal $x[n]=[-1.2, 0, 1, 2.5, 0.7, -0.1, -1.3, 0.1]$ is obtained by sampling $x(t)$ with sampling rate 800 samples per sec ($f_s=800$). Find 8 point DFT ($N=8$), $X[k]$. What is the magnitude of the signal at $F=200$ cycles/sec.	5M
2	Explain the theory of speech production. In your response, detail the physiological role of the a. vocal folds in determining phonation: voiced vs. unvoiced and pitch b. vocal tract (oral and nasal cavities) in shaping the signal via place and manner of articulation. Include example speech sounds for voiced and unvoiced, place and manner of articulation.	5M
3	Explain the algorithm to detect word boundaries in speech signals with all essential requirements so that one can implement it and apply for real world speech signals.	5M
4	What is a Spectrogram of speech signal and how is it computed? Given a speech signal of length 5 seconds, what is shape of spectrogram if Sampling rate 8KHz, Number of bins = 128, Hop size for framing = 10 ms Window length = 25 ms	5M

Solutions:

1. $N = \text{length of signal} = 8$
DFT of $x[n]$ is $X[k] = \sum_n x[n] e^{-j(2\pi/N)kn}$ with n and k both varying from 0 to 7

$$X[0] = 1.7$$

$$X[1] = -1.0206 - j2.2894$$

$$X[2] = -0.6 + j2.5$$

$$X[3] = -1.1794 + j1.0894$$

$$X[4] = 0.1$$

$$X[5] = -1.1794 - j1.0894$$

$$X[6] = -0.6 - j2.5$$

$$X[7] = -1.0206 + j2.2894$$

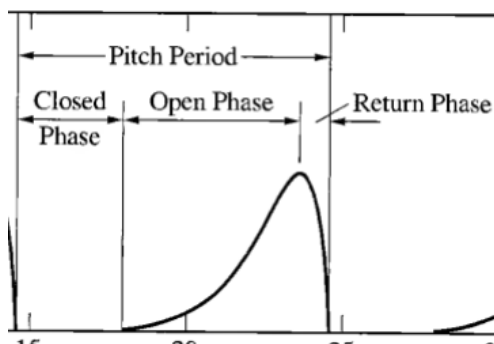
Frequency resolution in DFT is $f_s/N = 800/8 = 100$ Hz

$F = 200$ Hz corresponds to $k = F/\Delta f = 200/100 = 2$

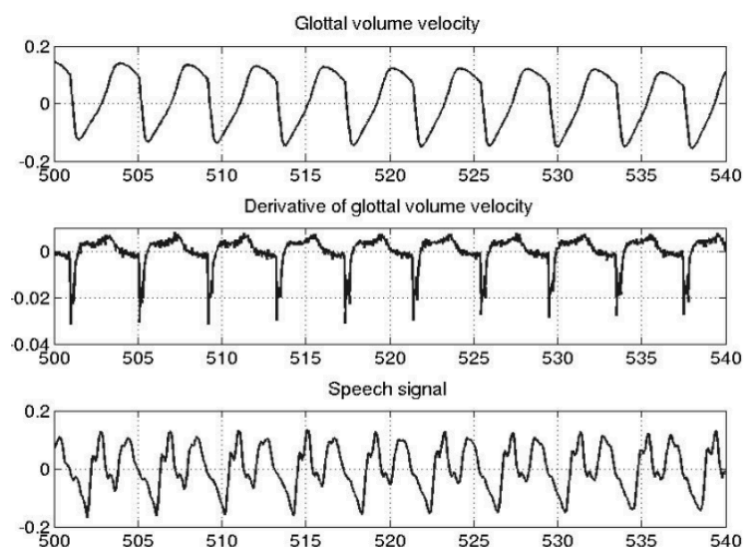
DFT magnitude at $k=2$ is $|X[2]| = |-0.6 + j2.5| = 2.57$

2. a. **Vocal folds:**

- These are muscular tissues located at the larynx.
- Vocal folds vibrate (open and close) when air flows from lungs while we speak. The sounds produced through vocal folds vibration are referred to as voiced speech. Ex: Vowels, /g/, /d/, /b/
- The air flow through glottis when we speak is shown as follows.



- One cycle of vocal fold vibration corresponds to one pitch period.
- The sounds produced by vocal tract without vocal folds vibrations are referred to as unvoiced speech. Ex: /s/, /h/, /k/, /t/



b. **Vocal tract:**

The vocal tract (oral cavity, nasal cavity, pharynx) acts as an acoustic filter, shaping glottal waveform through oral organs like tongue, jaw, nose etc., and creating formants (resonances).

Place of Articulation sounds produced based on where the constriction occurs):

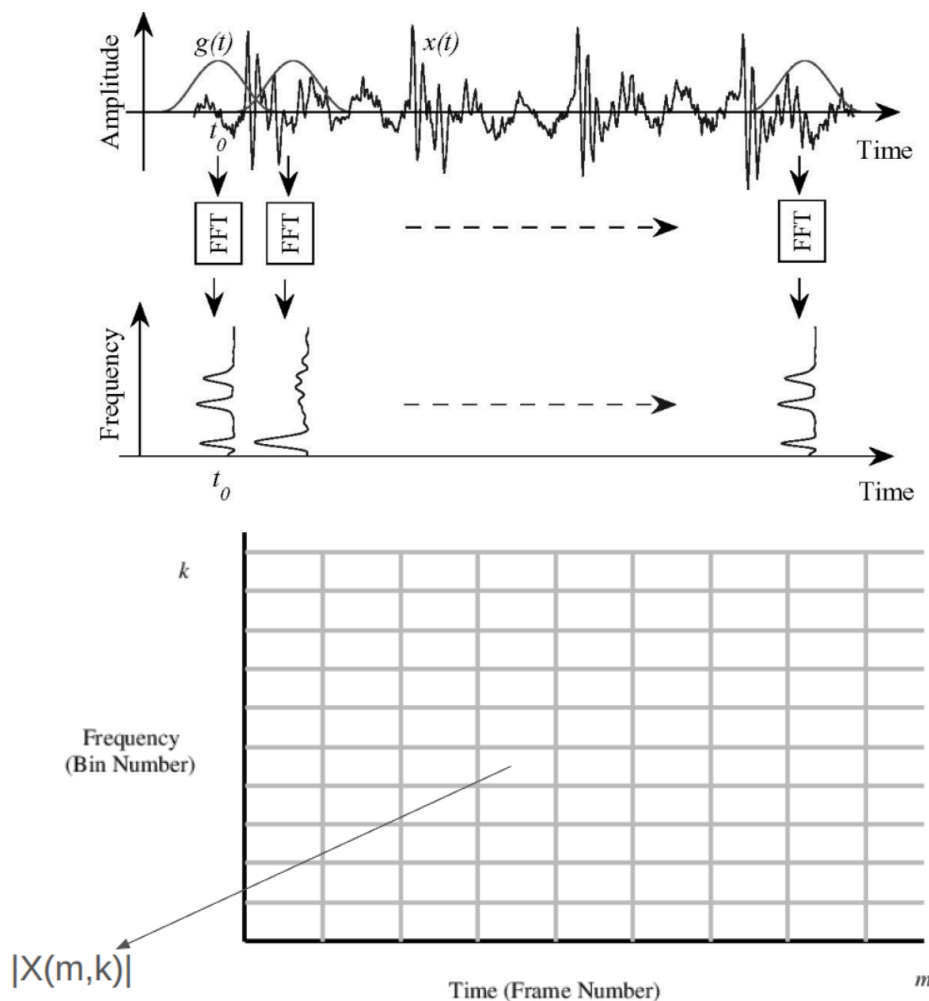
- Bilabial: Both lips (/p/, /b/, /m/)
- Labiodental: Lower lip vs upper teeth (/f/, /v/)
- Dental: Tongue vs teeth (/θ/ in "think", /ð/ in "this")
- Alveolar: Tongue vs alveolar ridge (/t/, /d/, /s/, /z/, /n/, /l/)
- Palatal: Tongue vs hard palate (/ʃ/ in "ship", /ʒ/ in "measure", /j/ in "yes")
- Velar: Tongue vs soft palate (/k/, /g/, /ŋ/ in "sing")
- Glottal: At vocal folds (/h/)

Manner of Articulation is based on how the air flows in oral and nasal cavities:

- Vowels: Open vocal tract with varying tongue position (/a/, /e/, /i/, /o/, /u/)
 - Stops/Plosives: Complete blockage then release (/p/, /b/, /t/, /d/, /k/, /g/)
 - Fricatives: Narrow constriction causing turbulence (/f/, /v/, /s/, /z/, /ʃ/, /ʒ/)
 - Affricates: Stop followed by fricative (/tʃ/ in "church", /dʒ/ in "judge")
 - Nasals: Air through nasal cavity (/m/, /n/, /ŋ/)
 - Liquids: Narrowing but no turbulence (/r/, /l/)
 - Glides: vowel like but with narrow passage (/w/, /j/)
3. Refer to lecture slides for energy and ZCR based word boundary detection or the reference paper by Rabiner and Sambur.
4. **Spectrogram**

A spectrogram is a 3D visual representation of the fourier spectrum (obtained by the short time fourier transform applied to speech frames.

) of frequencies in a signal as they vary with time. In general, it is the plot of the magnitude spectrum of a frame with x axis as frame number, y-axis as frequency bin number: $X[t,k]$.



Given:

f_s : 8000 Hz

Signal length: 5 seconds = $5 \times 8000 = 40000$ samples

DFT bins (N): 128

Hop size: 10 ms = $0.01 \times 8000 = 80$ samples

Window length: 25 ms = $0.025 \times 8000 = 200$ samples

Frame Rate = Sample Rate / Hop Length = $8000/80 = 100$ frames/sec

Number of frames for 5 secs = $5 \times 100 = 500$ frames

Spectrogram dimensions:

Time dimension: 500 frames

Frequency dimension (positive frequencies): $N/2 + 1 = 65$ bins

Spectrogram shape is : (65x500) matrix